

Introduction: What is an environmental problem?

Dr. Chen, Tingjia
School of Economics and Management
Harbin Institute of Technology (Shenzhen)

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Learning Objectives

After taking this class, you will be able to:

- *Get familiar with* each other and the course design
- *Understand* the characteristics of environmental goods
- *Illustrate* what "tragedy of the commons" is
- *Illustrate* why environmental problems occur
- *Acknowledge* the urgency to deal with environmental problems

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First thing first, let's get familiar with each other!

Please take several minutes to address the following questions:

- Who are you?
- What is your major?
- Why are you interested in this course? What do you expect to learn from course?

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Now, let's talk about how the course works!

Rationale of the Course Design

Evaluation System

class participation	Presentation	final report
30%	35%	35%

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Dimension	Excellent	Needs Improvement	Unacceptable
Relevance 10%	The presentation is clearly stated and deeply grounded in the topic/theories of this course. 8-10 pts	The presentation is vague or not tightly related to course's topic. 5-7 pts	There is barely related to the topic of this course. 0-4 pts
Evidence 10%	Case selected is highly relevant to the topic, is presented accurately and completely. 8-10 pts	Case selected is not relevant to the topic or is vague and incomplete – components are missing or inaccurate or unclear. 5-7 pts	There is barely evidence in the paper. Or, the evidence presented is irrelevant to the topic. 0-4 pts
Organization and clearance 15%	The presentation is coherently organized, and the logic is easy to follow. 11-15 pts	The presentation is generally well organized and most of the argument is easy to follow. 5-10 pts	The presentation lacks clarity and conciseness. It is hard for the audience to follow. 0-4 pts

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Grading Rubric of Final Paper for Using AI Chatbot

Dimension	Excellent	Good	Needs Improvement	Unacceptable
Research Purpose 10%	Research purpose is clearly stated. The purpose is deeply grounded in the topic/theories of this course and intends to address a practical relevant/real-world concerned issue. 8-10 pts	Research purpose is clear in parts. It has some relevance to this course. It addresses an issue with somewhat practical relevance. 5-7 pts	Research purpose is vague or not consistently maintained. It has weak relationship with the topics discussed in this course or have little practical implications . 2-4 pts	No research purpose is found in the paper. 0-1 pts
Capacity of Human–Computer Interaction 15%	Prompts are clearly built on human intellectual reflection and logic so that can effectively guide Conversational AI to generate relevant responses step by step. 12-15 pts	Prompts are somewhat built on human intellectual reflection and logic and generally effectively guide Conversational AI to generate relevant responses step by step. 7-11 pts	The logic of the prompts are vague or hard to follow. The responses generated by Conversational AI have weak relevance with the research purpose. 4-7 pts	There is barely logic in the prompts and cannot generate useful or relevant information in the AI responses. 0-3 pts
Outputs 5%	Final output is coherently organized, and the logic is easy to follow. 4-5 pts	The final output is generally well organized and most of the argument is easy to follow. 3 pts	Final output is poorly organized and difficult to read – does not flow logically from one part to another. 1-2 pts	Final output is hard to follow. 0-1 pts.
Evaluation and Reflection 5%	Provide solid and thorough evaluation on the accuracy, content, and quality of AI-generated output. 4-5 pts	Evaluation is mostly solid but may lack key components. 3 pts	Evaluation is biased and cannot provide concrete reasoning. 1-2 pts	Evaluation section is missing. 0-1 pts

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Remember!
I am here to help!

- Office: M315
- Office hour: by appointment
- Email: chentingjia@hit.edu.cn

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What are environmental goods?

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Characteristics of Environmental Goods

- Classification of Goods (物品的分类)

		竞争性 Subtractability	
		高竞争性 High subtractability	低竞争性 Low subtractability
排他性 Excludability	易排他 Easy to exclude	私人物品 Private goods (e.g., laptop, pen)	俱乐部物品 Club goods (e.g., university curriculum; database)
	不易排他 Hard to exclude	公共池塘资源 Common pool resources/sinks (e.g., park, road)	公共物品 Public goods (e.g., public course, policy)

- Excludability: whether or not a user can be efficiently excluded from using a resource. (一个人在为获取某一物品所付出代价时，在多大程度上可以将潜在受益人排除受益范围)
- Subtractability: the degree to which one person's use of a resource diminishes others' use (一个人所获取的物品/资源的数量在多大程度上影响他人对这种物品的收益)

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Class Activity I:
Identify the classification of the following goods

- Bottled water



- Water in the river



- Facilities in the gym



The classification of goods is conditional. We need to look into specific cases when conducting analysis.

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Tragedy of the Commons

TED Ed
LESSONS WORTH SHARING

Now, can you illustrate what the tragedy of the common?

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"Tragedy of the Commons" and Environmental Problems



- Environmental problems are usually embedded in common pool resources, which have the characteristics of low excludability and high subtractability.
- Environmental Problems:
  - High subtractability → overuse/over-discharge
  - Low excludability → "free rider"

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"Tragedy of the Commons" and Environmental Policy

- Assumption about human decision-making model in Hardin's "tragedy of the commons"
  - "Economic rational man"
  - Moral or conscience is not reliable.
- Hardin suggests to rely on government's coercion to solve the "tragedy of the commons". →i.e., command-and-control regulation
- Solutions to the "tragedy of the commons": technology and rules



Garrett Hardin (1915-2003)

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Class Activity II:

Please give me a real-life example about the "tragedy of commons".

Can you illustrate why environmental problem is usually a tragedy of commons?

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Externalities （外部性）

Externalities are also referred to as spillover effects (溢出效应).

In an economic system, the decisions and actions of an individual affect the benefits (or costs) of other individuals within the system.

It is an economic activity that imposes a negative effect on an unrelated third party. Pollution is termed an externality because it imposes costs on people who are "external" to the producer and consumer of the polluting product.

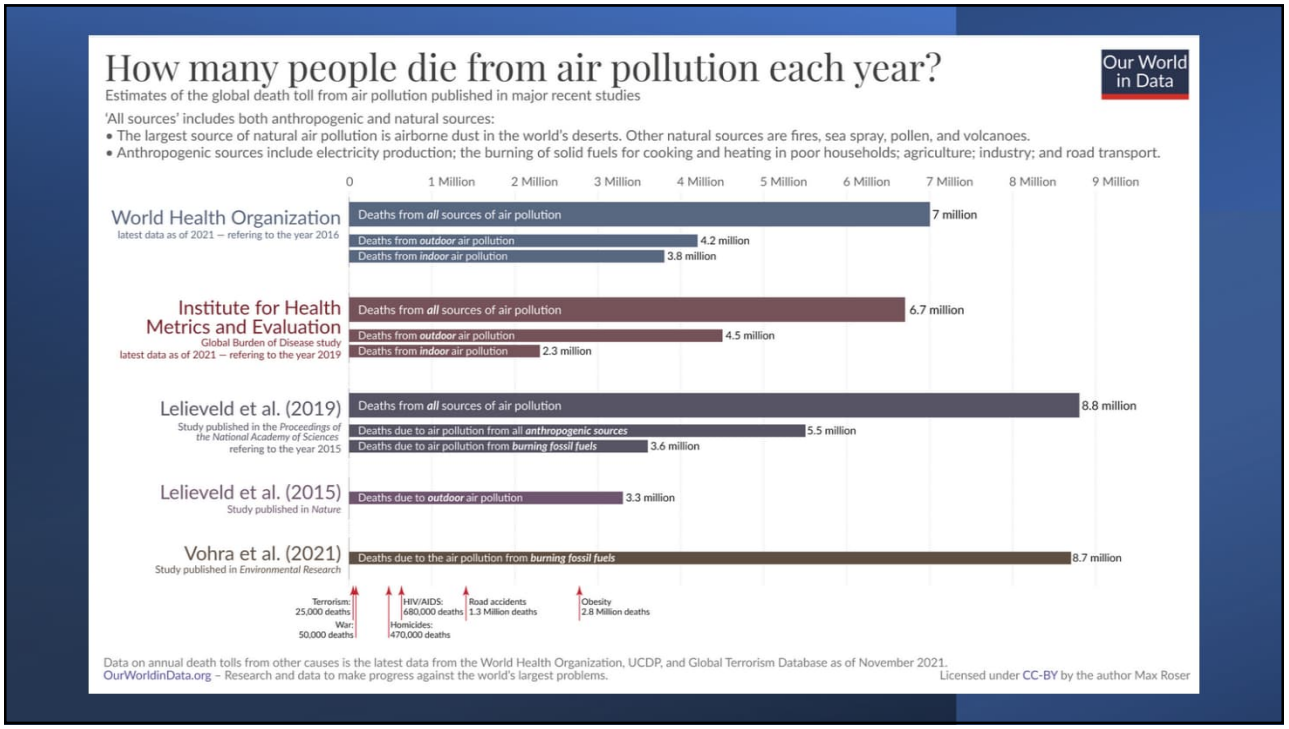
Positive impact of individual behavior on others within the system: Positive externality

Negative impact of individual behavior on others within the system: Negative externality



Can you provide some examples about externalities, positive or negative?

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